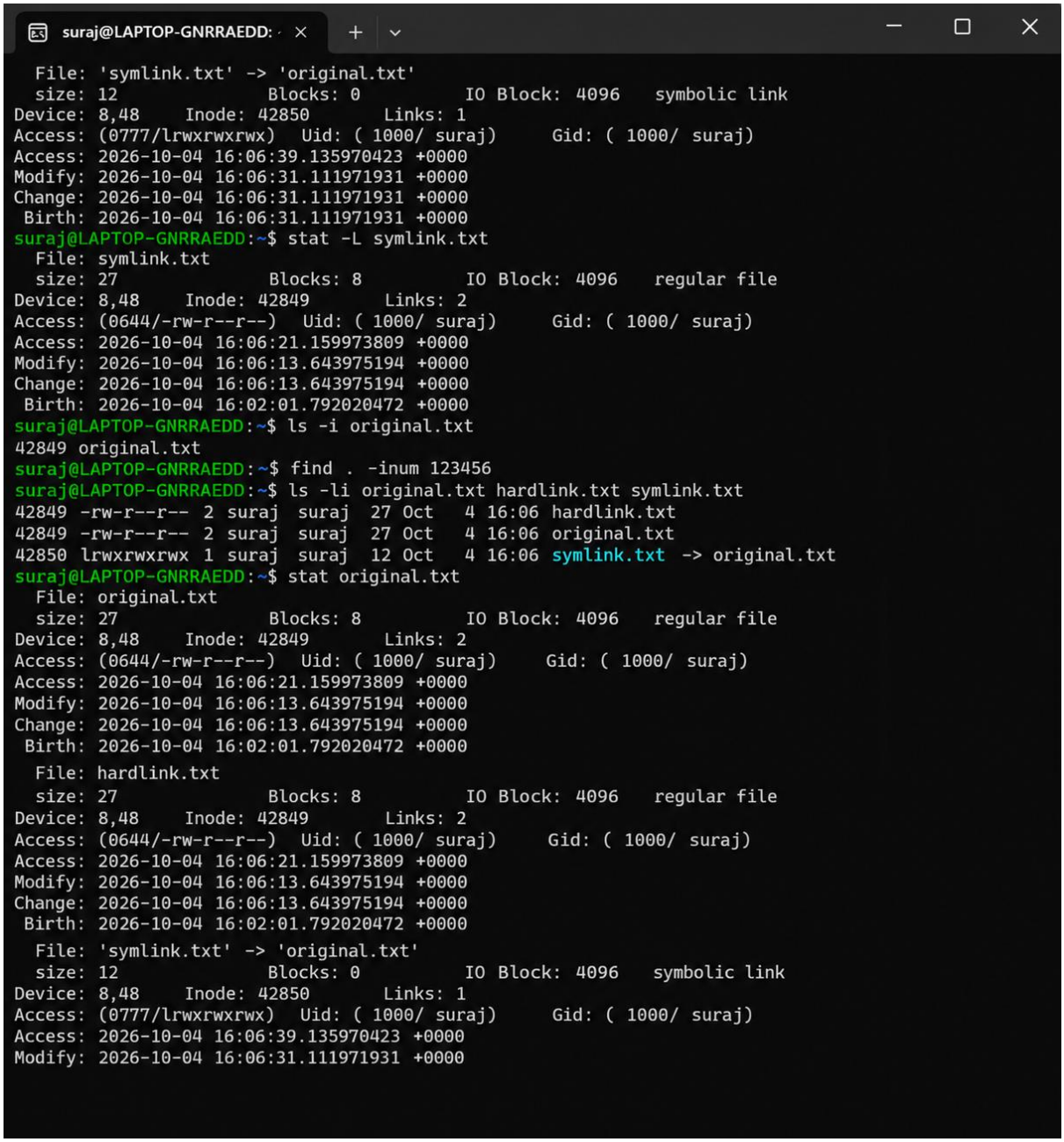
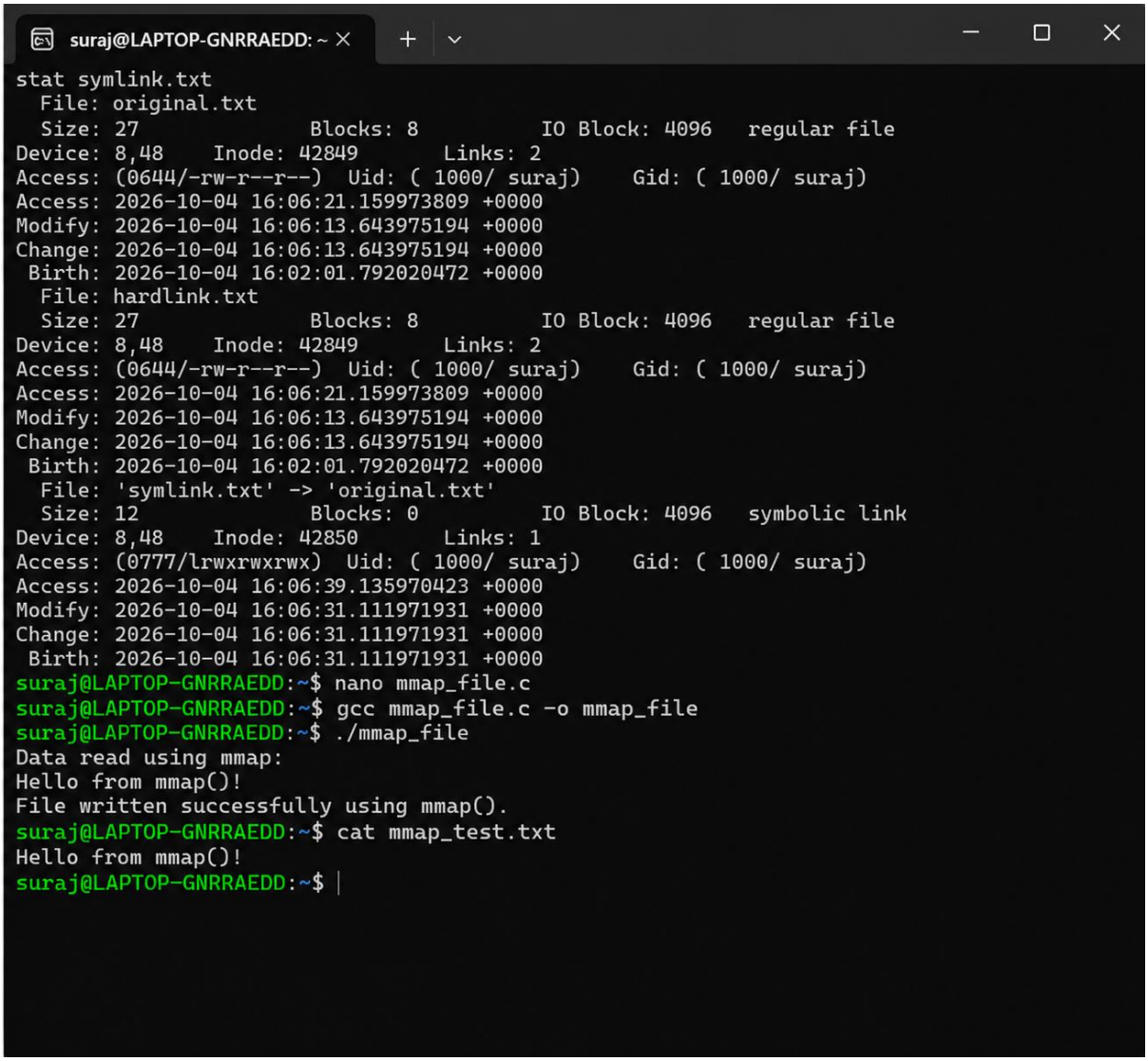
WEEK :-10

```
suraj@LAPTOP-GNRRAEDD:~$ echo "Hello Linux Inode" > original.txt
suraj@LAPTOP-GNRRAEDD:~$ cat original.txt
Hello Linux Inode
suraj@LAPTOP-GNRRAEDD:~$ ls -i original.txt
42849 original.txt
suraj@LAPTOP-GNRRAEDD:~$ 42849 > inode number
original.txt -> file nam
42849: command not found
original.txt: command not found
suraj@LAPTOP-GNRRAEDD:~$ stat original.txt
  File: original.txt
  Size: 18              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 42849       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ suraj)     Gid: ( 1000/ suraj)
Access: 2026-10-04 16:02:10.056018883 +0000
Modify: 2026-10-04 16:02:01.796020472 +0000
Change: 2026-10-04 16:02:01.796020472 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
suraj@LAPTOP-GNRRAEDD:~$ ln original.txt hardlink.txt
suraj@LAPTOP-GNRRAEDD:~$ ls -li original.txt hardlink.txt
42849 -rw-r--r-- 2 suraj suraj  18 Oct  4 16:02 hardlink.txt
42849 -rw-r--r-- 2 suraj suraj  18 Oct  4 16:02 original.txt
suraj@LAPTOP-GNRRAEDD:~$ echo "Modified through hard link" > hardlink.txt
suraj@LAPTOP-GNRRAEDD:~$ cat original.txt
Modified through hard link
suraj@LAPTOP-GNRRAEDD:~$ ln -s original.txt symlink.txt
suraj@LAPTOP-GNRRAEDD:~$ ls -li original.txt hardlink.txt symlink.txt
42849 -rw-r--r-- 2 suraj suraj  27 Oct  4 16:06 hardlink.txt
42849 -rw-r--r-- 2 suraj suraj  27 Oct  4 16:06 original.txt
42850 lrwxrwxrwx 1 suraj suraj  12 Oct  4 16:06 symlink.txt -> original.txt
suraj@LAPTOP-GNRRAEDD:~$ stat original.txt
  File: original.txt
  Size: 27              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 42849       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ suraj)     Gid: ( 1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
suraj@LAPTOP-GNRRAEDD:~$ stat original.txt
  File: original.txt
  Size: 27              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 42849       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ suraj)     Gid: ( 1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
suraj@LAPTOP-GNRRAEDD:~$ stat symlink.txt
```


suraj@LAPTOP-GNRRAEDD:
File: 'symlink.txt' -> 'original.txt'
size: 12 Blocks: 0 IO Block: 4096 symbolic link
Device: 8,48 Inode: 42850 Links: 1
Access: (0777/lrwxrwxrwx) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:39.135970423 +0000
Modify: 2026-10-04 16:06:31.111971931 +0000
Change: 2026-10-04 16:06:31.111971931 +0000
Birth: 2026-10-04 16:06:31.111971931 +0000
suraj@LAPTOP-GNRRAEDD:~$ stat -L symlink.txt
File: symlink.txt
size: 27 Blocks: 8 IO Block: 4096 regular file
Device: 8,48 Inode: 42849 Links: 2
Access: (0644/-rw-r--r--) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
Birth: 2026-10-04 16:02:01.792020472 +0000
suraj@LAPTOP-GNRRAEDD:~$ ls -i original.txt
42849 original.txt
suraj@LAPTOP-GNRRAEDD:~$ find . -inum 123456
suraj@LAPTOP-GNRRAEDD:~$ ls -li original.txt hardlink.txt symlink.txt
42849 -rw-r--r-- 2 suraj suraj 27 Oct 4 16:06 hardlink.txt
42849 -rw-r--r-- 2 suraj suraj 27 Oct 4 16:06 original.txt
42850 lrwxrwxrwx 1 suraj suraj 12 Oct 4 16:06 symlink.txt -> original.txt
suraj@LAPTOP-GNRRAEDD:~$ stat original.txt
File: original.txt
size: 27 Blocks: 8 IO Block: 4096 regular file
Device: 8,48 Inode: 42849 Links: 2
Access: (0644/-rw-r--r--) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
Birth: 2026-10-04 16:02:01.792020472 +0000
File: hardlink.txt
size: 27 Blocks: 8 IO Block: 4096 regular file
Device: 8,48 Inode: 42849 Links: 2
Access: (0644/-rw-r--r--) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
Birth: 2026-10-04 16:02:01.792020472 +0000
File: 'symlink.txt' -> 'original.txt'
size: 12 Blocks: 0 IO Block: 4096 symbolic link
Device: 8,48 Inode: 42850 Links: 1
Access: (0777/lrwxrwxrwx) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:39.135970423 +0000
Modify: 2026-10-04 16:06:31.111971931 +0000


suraj@LAPTOP-GNRRAEDD: ~
stat symlink.txt
File: original.txt
Size: 27 Blocks: 8 IO Block: 4096 regular file
Device: 8,48 Inode: 42849 Links: 2
Access: (0644/-rw-r--r--) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
Birth: 2026-10-04 16:02:01.792020472 +0000
File: hardlink.txt
Size: 27 Blocks: 8 IO Block: 4096 regular file
Device: 8,48 Inode: 42849 Links: 2
Access: (0644/-rw-r--r--) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
Birth: 2026-10-04 16:02:01.792020472 +0000
File: 'symlink.txt' -> 'original.txt'
Size: 12 Blocks: 0 IO Block: 4096 symbolic link
Device: 8,48 Inode: 42850 Links: 1
Access: (0777/lrwxrwxrwx) Uid: (1000/ suraj) Gid: (1000/ suraj)
Access: 2026-10-04 16:06:39.135970423 +0000
Modify: 2026-10-04 16:06:31.111971931 +0000
Change: 2026-10-04 16:06:31.111971931 +0000
Birth: 2026-10-04 16:06:31.111971931 +0000
suraj@LAPTOP-GNRRAEDD:~$ nano mmap_file.c
suraj@LAPTOP-GNRRAEDD:~$ gcc mmap_file.c -o mmap_file
suraj@LAPTOP-GNRRAEDD:~$ ./mmap_file
Data read using mmap:
Hello from mmap()!
File written successfully using mmap().
suraj@LAPTOP-GNRRAEDD:~$ cat mmap_test.txt
Hello from mmap()!
suraj@LAPTOP-GNRRAEDD:~$